

LQR for Fault Accommodation, Analyzed Using Analytical $\mathcal{H}_2$ -Norm

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1 Introduction

The purpose of this document is to explore the link between full-state feedback and LQR in the context of fault accommodation using the method introduced by Allen and El-Farra in their publication focused on calculating an accommodating fault parameter $\hat{\theta}_j$ to obtain a controller that is both stable and high-performing in terms of the $\mathcal{H}_2$ -norm [1].

It is hypothesized that, since full-state feedback and LQR use the same state-space model and linear control law just with different gain matrices K_{FS} and K_{LQR} , the calculation of an accommodating $\hat{\theta}_j$ chosen such that the $\mathcal{H}_2$ -norm is minimized will yield a controller that is equivalent to LQR, regardless of the initial choice of K_{FS} or K_{LQR} . This draws on the fact that LQR, at least partially, already has a notion of $\mathcal{H}_2$ -norm minimization by virtue of it minimizing a quadratic cost function.

2 The Model

2.1 Non-Isothermal CSTR

The system we will be applying fault accommodation to is a simple two-state, non-isothermal CSTR with a heating jacket containing the reaction $A \rightarrow B$:

$$\frac{dC_A}{dt} = \frac{F_{A0}}{V}(C_{A0} - C_A) - A_0 e^{-E/RT} C_A \quad (1)$$

$$\frac{dT}{dt} = \frac{F_{A0}}{V}(T_0 - T) + \frac{(-\Delta H)}{\rho C_p} A_0 e^{-E/RT} C_A - \frac{UA}{\rho C_p V}(T - T_c) \quad (2)$$

The manipulated variable is the inlet volumetric flow rate of A , F_{A0} . For the purposes of applying the theory of Allen and El-Farra's paper, the model will be linearized to obtain the A and B matrices,

$$A = \begin{bmatrix} -\left(\frac{F_{A0}^{eq}}{V} + A_0 e^{-E/RT_{eq}}\right) & \frac{A_0 E}{RT_{eq}^2} e^{-E/RT_{eq}} C_A^{eq} \\ \frac{(-\Delta H)}{\rho C_p} & \left(\frac{A_0 E}{RT_{eq}^2} e^{-E/RT_{eq}} C_A^{eq} - \frac{F_{A0}^{eq}}{V} - \frac{UA}{\rho C_p V}\right) \end{bmatrix}, \quad B = \begin{bmatrix} \frac{C_{A0} - C_A^{eq}}{V} \\ \frac{T_0 - T_{eq}}{V} \end{bmatrix}. \quad (3)$$

The state-space model takes the form,

$$\dot{x} = Ax + B\theta u, \quad (4)$$

$$\dot{\hat{x}} = \hat{A}\hat{x} + \hat{B}\hat{\theta}u, \quad (5)$$

with the linear control law,

$$u = -K\hat{x}. \quad (6)$$

$\theta \in \mathbb{R}^{m \times m}$ is the true fault parameter and $\hat{\theta} \in \mathbb{R}^{m \times m}$ is the estimated fault parameter. They are square diagonal matrices containing the multiplicative faults for each actuator (in our case, $m = 1$, so it is a scalar).

2.2 Estimation Error and Augmented State Vector

Following the steps from Richard M. Murray's lecture notes on state estimation [2], we define the error between the state and estimation as $e = x - \hat{x}$, and an augmented state vector to take this error into account, $\xi = [x^\top \ e^\top]^\top$. This means that, using the control law in equation (6), we have,

$$\dot{\xi} = \begin{bmatrix} \dot{x} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} Ax - B\theta K\hat{x} \\ Ax - B\theta K\hat{x} - \hat{A}\hat{x} + \hat{B}\hat{\theta}K\hat{x} \end{bmatrix} \quad (7)$$

Substituting $\hat{x} = x - e$ which we get from our definition of e , we have,

$$\begin{bmatrix} \dot{x} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} Ax - B\theta K(x - e) \\ Ax - B\theta K(x - e) - \hat{A}(x - e) + \hat{B}\hat{\theta}K(x - e) \end{bmatrix}. \quad (8)$$

After rearrangement, we have,

$$\dot{\xi} = \Lambda \xi, \quad (9)$$

where,

$$\Lambda = \begin{bmatrix} A - B\theta K & B\theta K \\ (A - \hat{A}) - (B\theta - \hat{B}\hat{\theta})K & \hat{A} + (B\theta - \hat{B}\hat{\theta})K \end{bmatrix}. \quad (10)$$

2.3 State-Space with Disturbance

We need to consider what happens when the system undergoes an impulse disturbance, thus we need to describe the disturbance response of the system,

$$\dot{x} = Ax + Bu + Ed \quad (11)$$

where $d \in \mathbb{R}^q$ is the disturbance vector and $E \in \mathbb{R}^{n \times q}$ is the disturbance matrix. We will choose to apply an

impulse disturbance in the temperature, which allows us to determine E through the Jacobian,

$$E = \begin{bmatrix} \frac{A_0 E}{RT_{eq}^2} e^{-E/RT_{eq}} C_A^{eq} \\ \left(\frac{A_0 E}{RT_{eq}^2} e^{-E/RT_{eq}} C_A^{eq} - \frac{F_{A0}^{eq}}{V} - \frac{UA}{\rho C_p V} \right) \end{bmatrix} \quad (12)$$

We also need to define a performance output $z \in \mathbb{R}^a$, which is just a linear mapping of the states,

$$z = Qx \quad (13)$$

where $Q \in \mathbb{R}^{a \times n}$ is a weighting matrix. With this, we can formulate the augmented state-space model,

$$\dot{\xi} = \Lambda \xi + Hd \quad (14)$$

$$z = N\xi \quad (15)$$

where $H = [E^\top \ E^\top]^\top$ and $N = [Q \ 0]$.

2.4 $\mathcal{H}_2$ -Norm

We are finally able to define the (extended) $\mathcal{H}_2$ -norm of the impulse response of our closed-loop system:

$$\|G\|_{\mathcal{H}_2} = \text{trace} \left(H^\top X H \right)^{1/2} \quad (16)$$

where X is the solution to the Lyapunov equation,

$$M^\top X M - X + W_0 = 0 \quad (17)$$

where $M = I_0 e^{\Lambda \Delta}$ and $W_0 = \int_0^\Delta e^{\Lambda^\top t} N^\top N e^{\Lambda t} dt$. Lastly,

$$I_0 = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}. \quad (18)$$

For a stable system, we require that $\lambda_{max}(M) < 1$, which can help us prune our results.

3 Full-State Feedback

The full-state feedback controller was tuned such that the closed-loop poles were placed at $s_1 = -0.1$ and $s_2 = -0.2$, resulting in $K_{FS} = [46.886 \ 0.077]$. Calculations were done using the `control.place()` command from the Python `control` library.

4 Linear-Quadratic Regulator (LQR)

The LQR controller resulted in the gain $K_{LQR} = [0.525 \ -0.196]$, where the closed-loop poles were located at $s_1 = -0.0054$ and $s_2 = -0.233$. Calculations were done using the `control.lqr()` command from the Python `control` library.

5 Results and Discussion

The results of the $\mathcal{H}_2$ -norm depends heavily on the sampling time Δ . A shorter sampling time typically results in an $\mathcal{H}_2$ -norm surface that shows that, surprisingly, any choice of $\hat{\theta}$ results in the same norm. Figure (1) below shows both the LQR and full-state feedback norm contours.

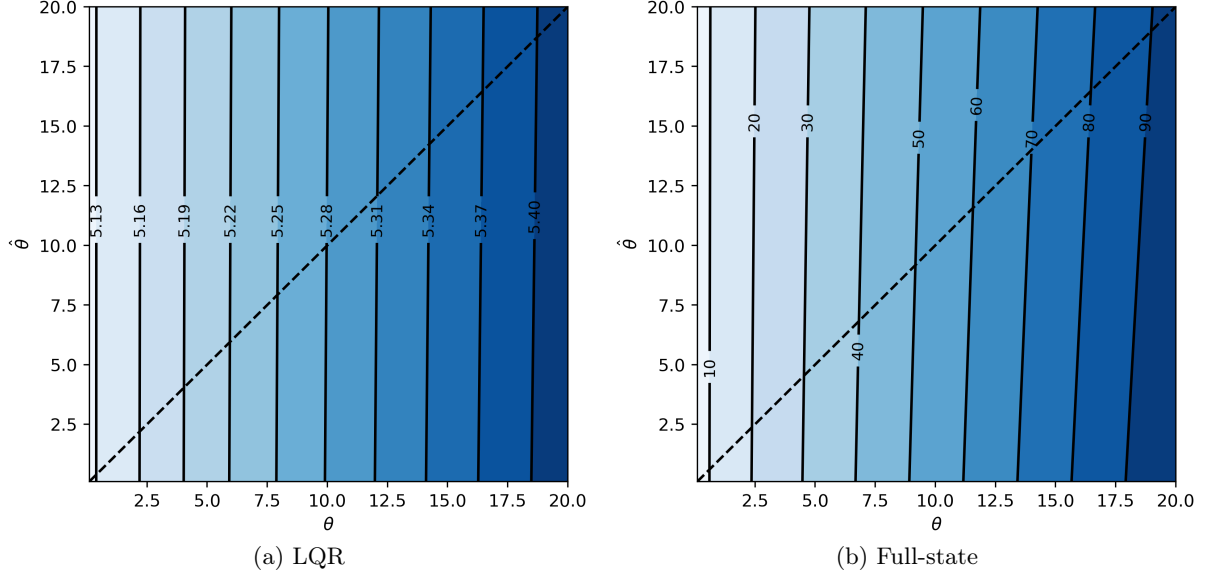


Figure 1: Both the LQR and full-state feedback norm contours show a similar $\hat{\theta}$ - θ dependence, where it seems that at any θ -value, any choice of $\hat{\theta}$ will result in the same norm.

We can verify that the dynamic behavior of systems with small Δ -values are similar by simulation; figure (2) shows the dynamic behavior of the concentration and temperature profiles with varying $\hat{\theta}$ - θ configurations.

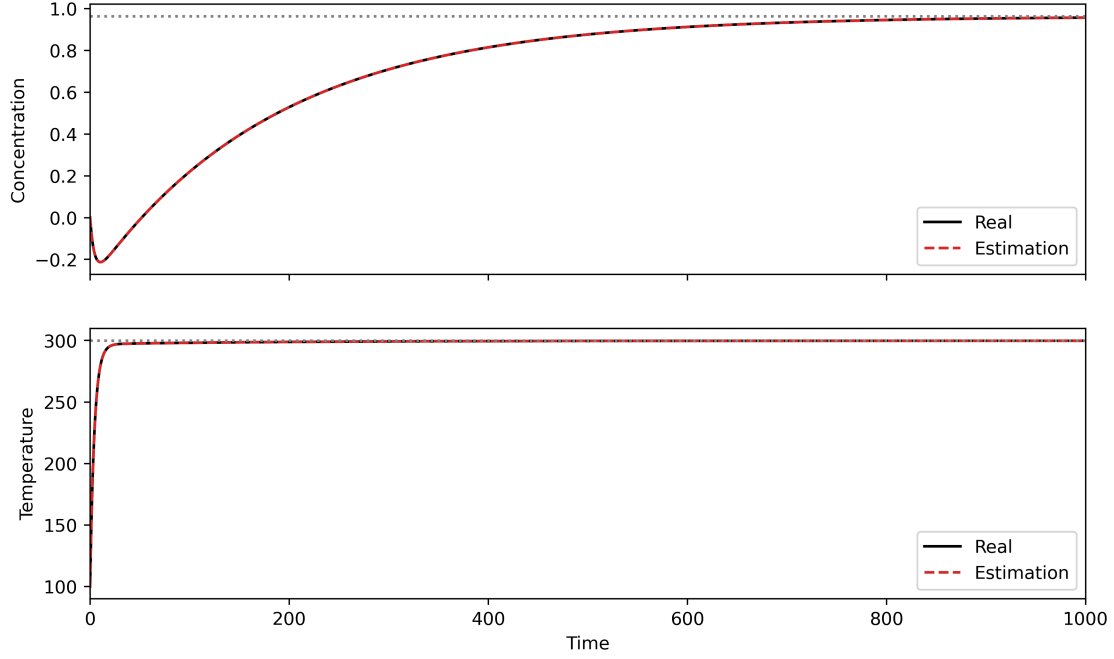
The results become more interesting when Δ becomes larger, as this is where the $\mathcal{H}_2$ -norms for LQR and full-state feedback start to show fundamental differences. Surprisingly, the norm contours for LQR show that there really are not unstable configurations, no matter the Δ -value. Figure (3) shows multiple contours, each with varying Δ .

What is interesting with the LQR norm contours is that between $\Delta = 50$ and $\Delta = 100$, the shape of the contours seems to switch; for $\Delta = 50$, the norm increases top-to-bottom, left-to-right, however for $\Delta = 100$, the norm now increases bottom-to-top, right-to-left. This hints that there must be some Δ where this switch occurs, where there must be a valley of sorts that contains minima. There is no saying that those minima are the best out of all possible configurations for Δ , but it is still an interesting thought.

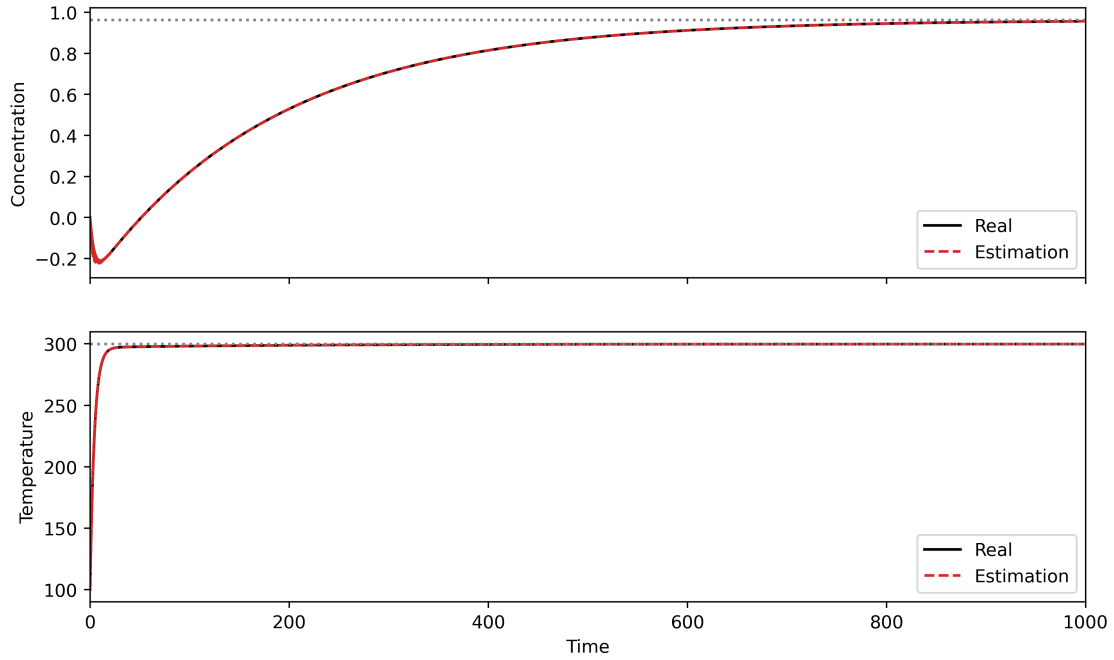
Moving away from LQR, we analyze the norm contours for full-state feedback. Here, we see that the norm is much more sensitive to changes in Δ than is LQR. A major shift in the contours occurs between $\Delta = 1$ and $\Delta = 5$, where a sort of unstable region in the bottom right corner of the plot forms, with small islands of relative stability forming on the border. This region is made more clear when we go to $\Delta = 10$, where it seems that the region approximately exists where $\hat{\theta} \leq \theta/2$.

Like what we did with LQR, we can verify that these regions do indeed produce stable or unstable state trajectories. Figure (5) shows a few sample trajectories at $\Delta = 5$.

Figure (6) shows some more sample state trajectories, because they look pretty cool. For these, $\Delta = 10$.



(a) LQR, $\theta = 1$, $\hat{\theta} = 1$.



(b) LQR, $\theta = 1$, $\hat{\theta} = 10$.

Figure 2: The dynamic response of the system is nearly identical, regardless of the choice of $\hat{\theta}$. $\Delta = 0.1$. The full-state feedback response is identical at this sampling rate.

6 Conclusions

It is still yet unclear as to what these results mean, however we will keep them in mind as we get closer to defining a problem and working towards at least a partial solution. After deliberating on these results for

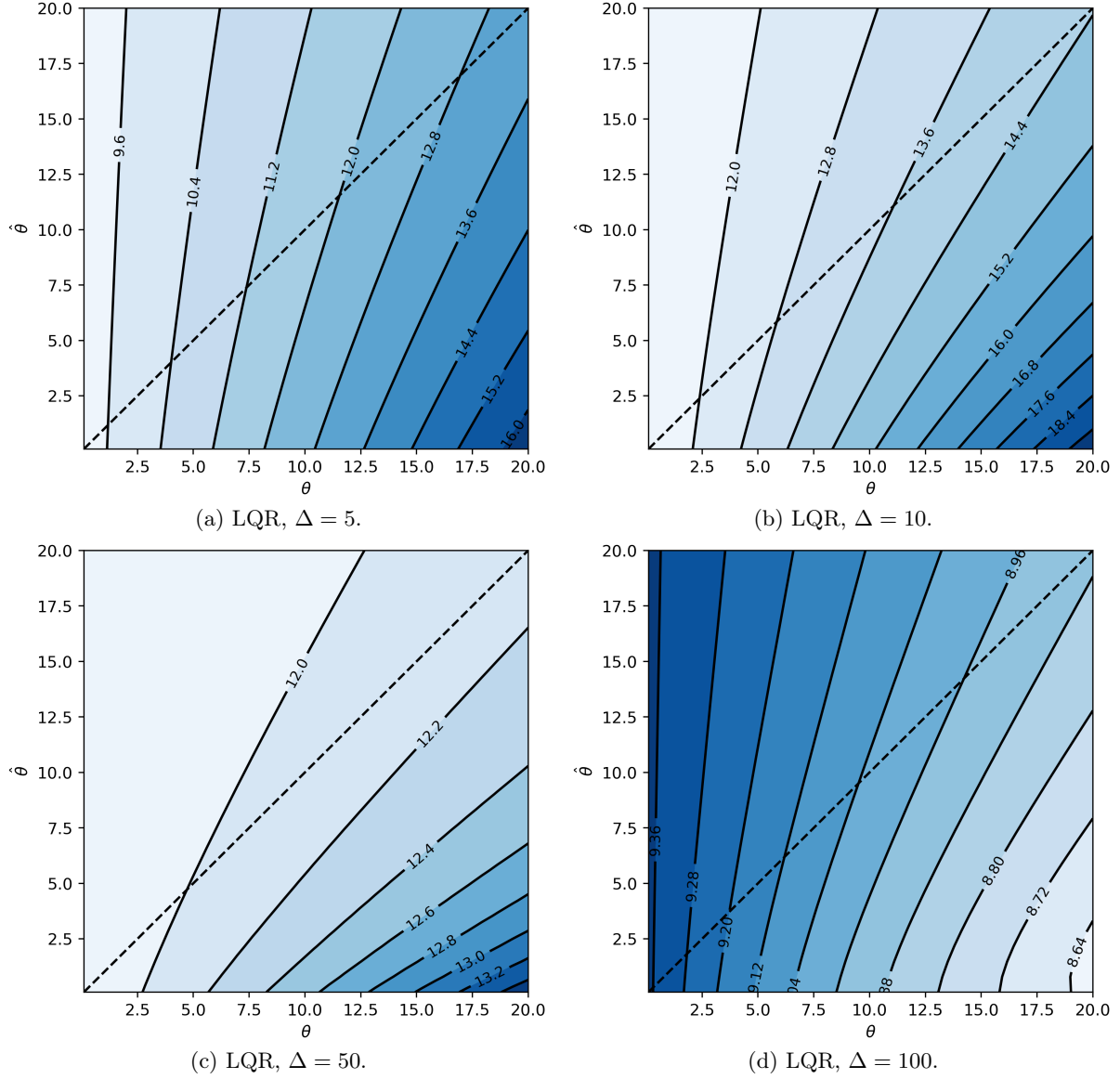
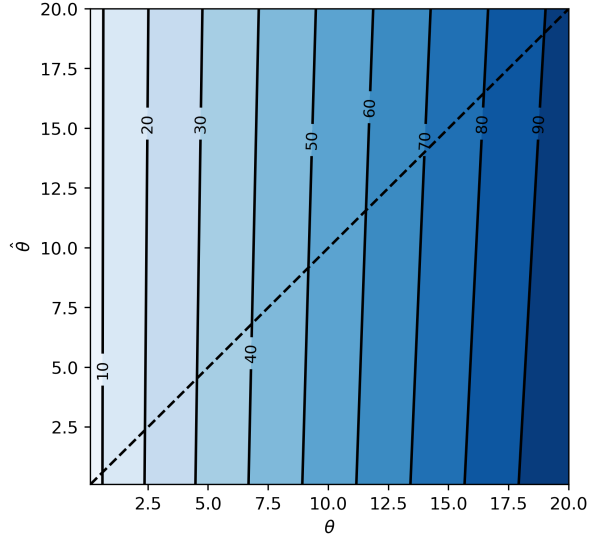
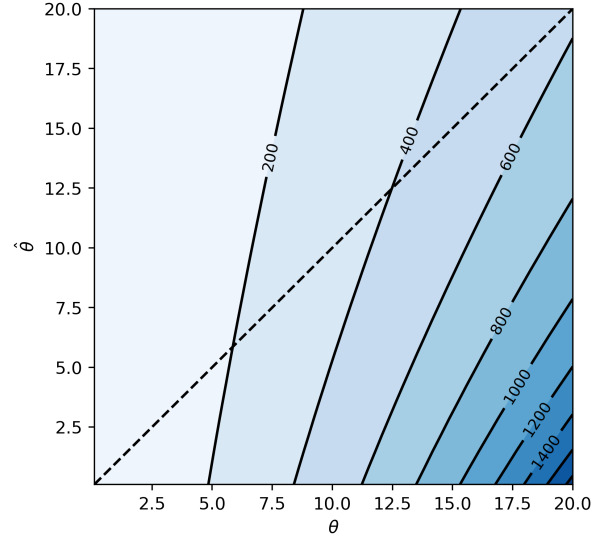


Figure 3: $\mathcal{H}_2$ -norm contours, LQR.

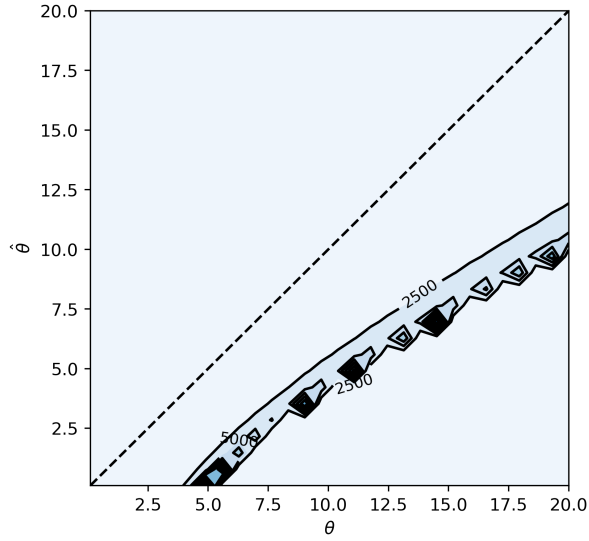
the better part of 2 hours, Professor El-Farra and I have narrowed down a possible route to take for the research project: perhaps LQR can be used to inform the MPC algorithm about regions within parameter space that yield a state trajectory with prescribed performance values, based on the $\mathcal{H}_2$ -norm. These regions can be passed to the MPC algorithm through the constraints, perhaps through defining upper and lower bounds for the control input. The details still need to be ironed out, though at least there's something at least somewhat promising.



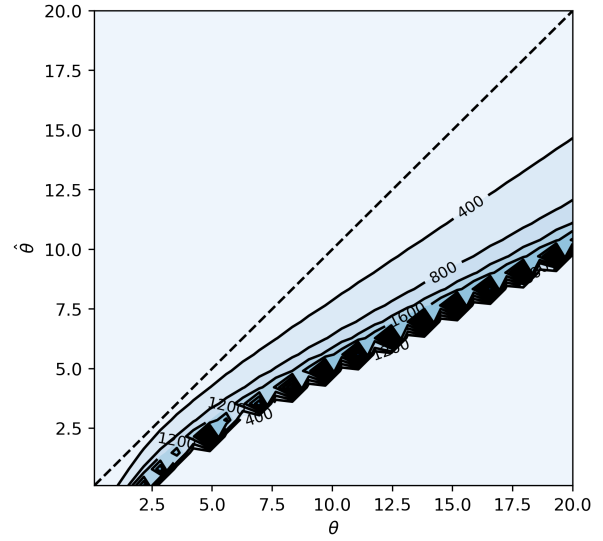
(a) Full-state, $\Delta = 0.1$.



(b) Full-state, $\Delta = 1$.

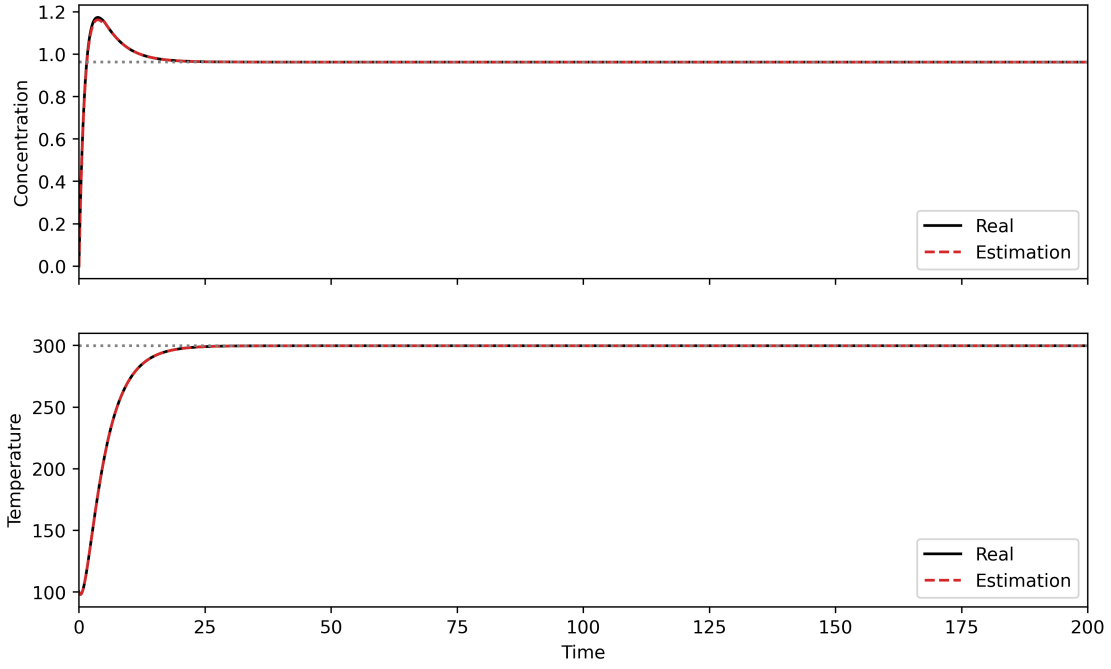


(c) Full-state, $\Delta = 5$.

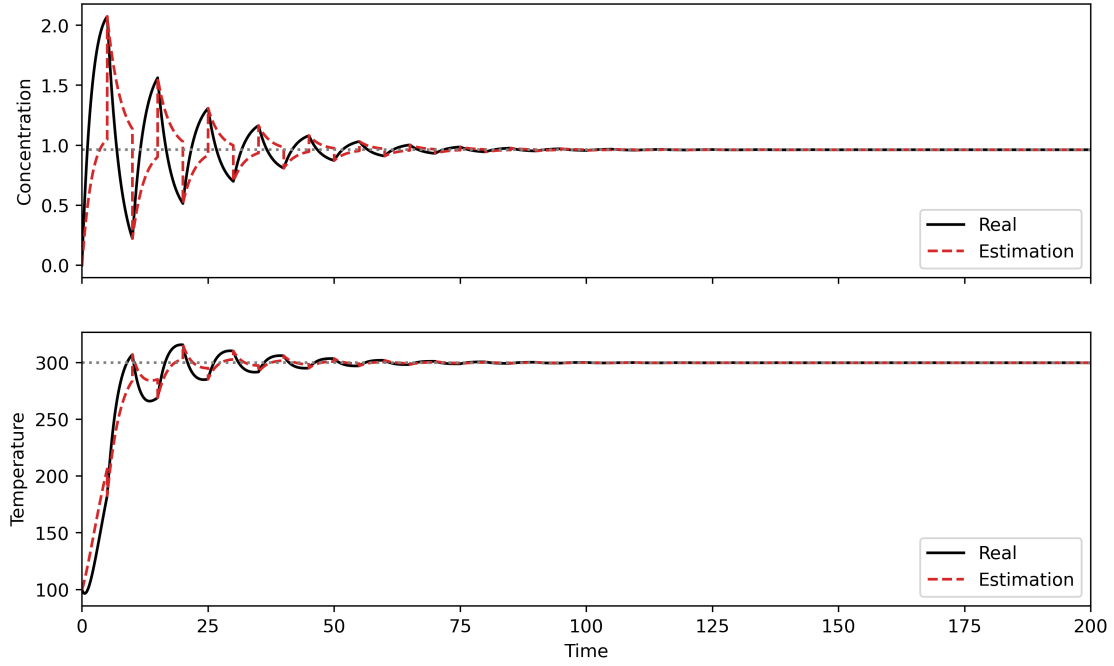


(d) Full-state, $\Delta = 10$.

Figure 4: $\mathcal{H}_2$ -norm contours, full-state feedback.

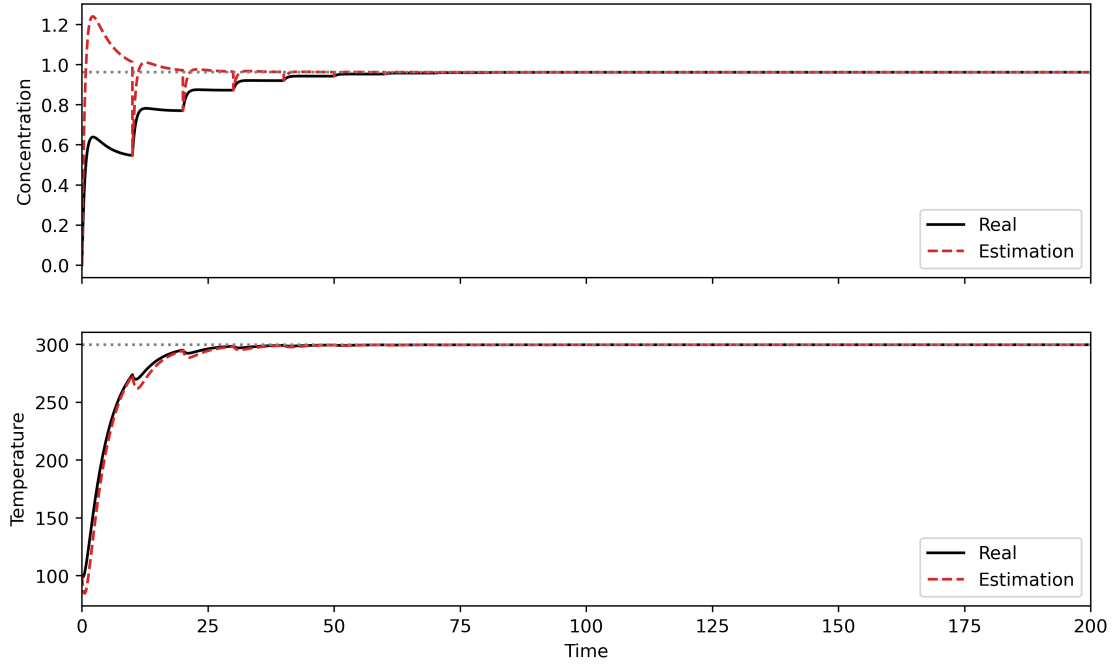


(a) Full-state feedback, $\theta = 10$, $\hat{\theta} = 10$.

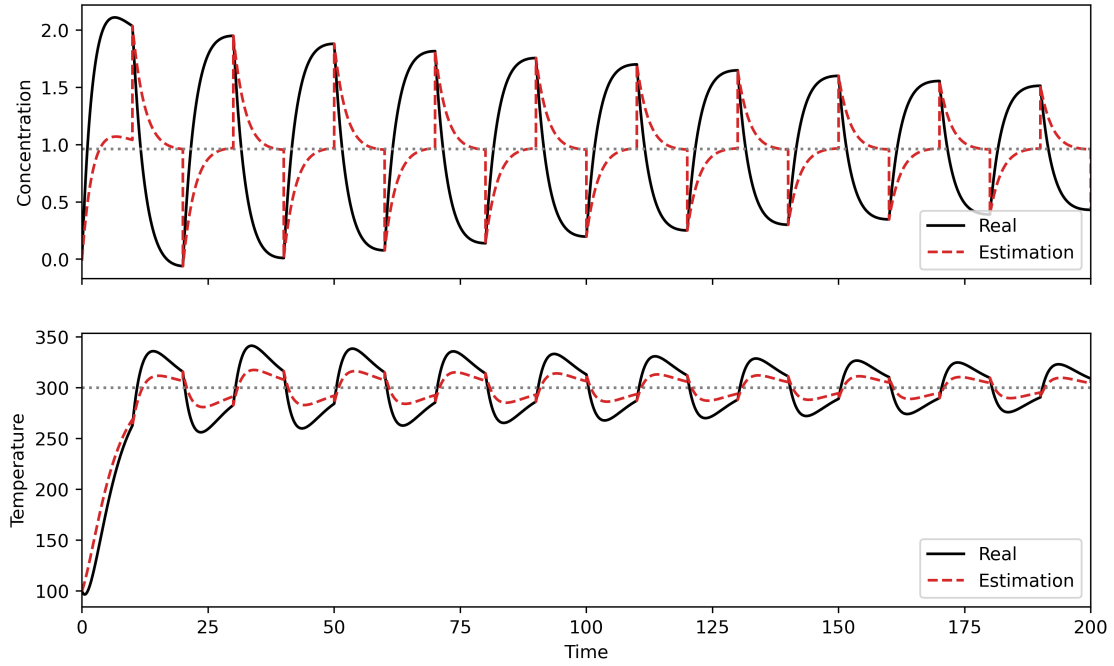


(b) Full-state feedback, $\theta = 10$, $\hat{\theta} = 5$.

Figure 5: The dynamic response of the system is strongly affected by a change in $\hat{\theta}$ - θ configuration with larger Δ . $\Delta = 5$.



(a) Full-state feedback, $\theta = 10$, $\hat{\theta} = 20$.



(b) Full-state feedback, $\theta = 10$, $\hat{\theta} = 5$.

Figure 6: The dynamic response of the system with very large Δ looks pretty cool. $\Delta = 10$.

References

- [1] J. T. Allen and N. H. El-Farra, “A model-based framework for fault estimation and accommodation applied to distributed energy resources,” *Renewable Energy*, vol. 100, p. 35–43, Jan 2017.
- [2] R. M. Murray, “State and output feedback.” https://www.cds.caltech.edu/~murray/courses/cds101/fa04/caltech/am04_ch5-24oct04.pdf, 2024.